Q1 - 24 January - Shift 1

Let a tangent to the curve $y^2 = 24x$ meet the curve $xy = 2$ at the points A and B. Then the mid points of such line segments AB lie on a parabola with the

(1) directrix $4x = 3$
(2) directrix $4x = -3$
(3) Length of latus rectum $\frac{3}{2}$
(4) Length of latus rectum 2

Space for your notes:

Q2 - 25 January - Shift 1

For some $a, b, c \in \mathbb{N}$, let $f(x) = ax - 3$ and

$g(x) = x^b + c, x \in \mathbb{R}$. If $(fog)^{-1}(x) = \left(\frac{x-7}{2}\right)^{1/3}$

then (fog) (ac) + (gof) (b) is equal to ____.

Space for your notes:

Q3 - 25 January - Shift 2

Let T and C respectively be the transverse and conjugate axes of the hyperbola $16x^2 - y^2 + 64x + 4y + 44 = 0$. Then the area of the region above the parabola $x^2 = y + 4$, below the transverse axis T and on the right of the conjugate axis C is:

(1) $4\sqrt{6} + \frac{44}{3}$
(2) $4\sqrt{6} + \frac{28}{3}$
(3) $4\sqrt{6} - \frac{44}{3}$
(4) $4\sqrt{6} - \frac{28}{3}$

Space for your notes:

Q4 - 31 January - Shift 2

Let H be the hyperbola, whose foci are $(1 \pm \sqrt{2}, 0)$ and eccentricity is $\sqrt{2}$. Then the length of its latus rectum is ______________.

(1) 2 (2) 3

(3) $\frac{5}{2}$ (4) $\frac{3}{2}$

Space for your notes:

Q5 - 01 February - Shift 2

Let $P(x_0, y_0)$ be the point on the hyperbola $3x^2 - 4y^2 = 36$, which is nearest to the line $3x + 2y = 1$. Then $\sqrt{2}(y_0 - x_0)$ is equal to :

(1) –3

(2) 9

(3) –9

(4) 3

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (1) **Q2** (2039) **Q3** (2) **Q4** (1)

Q5 (3)

Q1 (1)

$y^2 = 24x$

$a = 6$

$xy = 2$

$AB \equiv ty = x + 6t^2$ ....(1)

$AB \equiv T = S_1$

$kx + hy = 2hk$ .... (2)

From (1) and (2)

$\frac{k}{1} = \frac{h}{-t} = \frac{2hk}{-6t^2}$

$\Rightarrow$ then locus is $y^2 = -3x$

Therefore directrix is $4x = 3$

Q2 (2039)

Let $fog(x) = h(x)$

$$\Rightarrow h^{-1}(x) = \left(\frac{x-7}{2}\right)^{\frac{1}{3}}$$

$$\Rightarrow h(x) = fog(x) = 2x^3 + 7$$

$$fog(x) = a(x^b + c) - 3$$

$$\Rightarrow a = 2, b = 3, c = 5$$

$$\Rightarrow fog(ac) = fog(10) = 2007$$

$$g(f(x) = (2x-3)^3 + 5$$

$$\Rightarrow gof(b) = gof(3) = 32$$

$$\Rightarrow sum = 2039$$

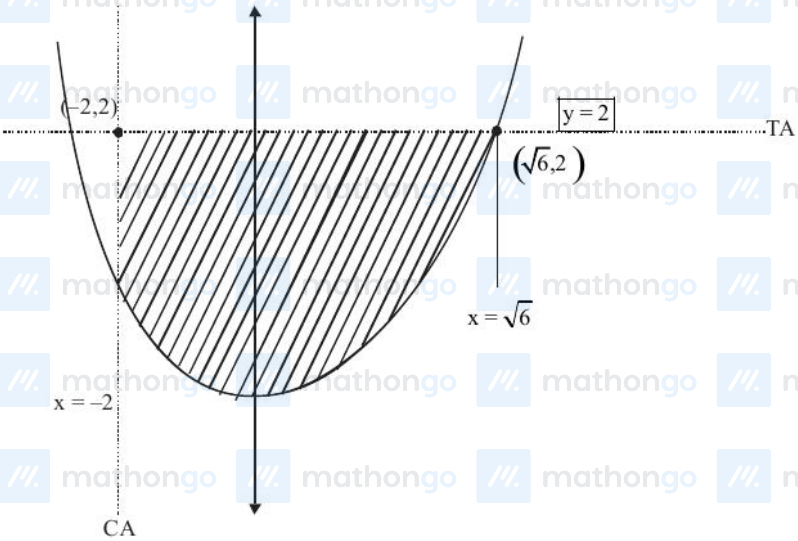
Q3 (2)

$$16(x^2+4x)-(y^2-4y)+44=0$$

$$16(x+2)^2-64-(y-2)^2+4+44=0$$

$$16(x+2)^2-(y-2)^2=16$$

$$\frac{(x+2)^2}{1}-\frac{(y-2)^2}{16}=1$$



$$A=\int_{-2}^{\sqrt{6}}\left(2-\left(x^2-4\right)\right)dx$$

$$A=\int_{-2}^{\sqrt{6}}\left(6-x^2\right)dx=\left(6x-\frac{x^3}{3}\right)_{-2}^{\sqrt{6}}$$

$$A=\left(6\sqrt{6}-\frac{6\sqrt{6}}{3}\right)-\left(-12+\frac{8}{3}\right)$$

$$A=\frac{12\sqrt{6}}{3}+\frac{28}{3}$$

$$A=4\sqrt{6}+\frac{28}{3}$$

Q4 (1)

$$2ae = \left|(1+\sqrt{2})-(1+\sqrt{2})\right| = 2\sqrt{2}$$

$$ae = \sqrt{2}$$

$$a = 1$$

$\Rightarrow b = 1 \quad \because e = \sqrt{2} \Rightarrow$ Hyperbola is rectangular

$$\Rightarrow L.R = \frac{2b^2}{a} = 2$$

Q5 (3)

$$3x^2 - 4y^2 = 36 \qquad 3x + 2y = 1$$

$$m = -\frac{3}{2}$$

$$m = +\frac{\sec\theta \; 3}{\sqrt{12}\,.\tan\theta}$$

$$\Rightarrow \frac{3}{\sqrt{12}} \times \frac{1}{\sin\theta} = \frac{-3}{2}$$

$$\sin\theta = -\frac{1}{\sqrt{3}}$$

$$\left(\sqrt{12}\,.\sec\theta,\ 3\tan\theta\right)$$

$$\left(\sqrt{12}\,.\frac{\sqrt{3}}{\sqrt{2}}, -3\times\frac{1}{\sqrt{2}}\right) \Rightarrow \left(\frac{6}{\sqrt{2}}, \frac{-3}{\sqrt{2}}\right)$$